

# Practice Task: E-Commerce Product Catalog Display

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## Project Overview

Build a functional E-Commerce Product Catalog Display page

## Application Requirements

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### Part 1: Project Setup (15 minutes)

#### Setup Tasks:

1. Create a new Angular project named `product-catalog`
2. Set up your development environment (VS Code, Angular CLI)
3. Import `FormsModule` in your `app.module.ts` for two-way binding
4. Create the basic component structure (see Component Architecture below)

```
ng new product-catalog
cd product-catalog
ng serve
```

#### Components to Create:

```
ng g c product-list
```

## Part 3: Product List Component (45 minutes)

### Required Features:

#### 3.1 Product Data Structure

Create an array of products with the following properties:

```
// product-list.component.ts
products = [
  {
    id: 1,
    name: 'Wireless Headphones',
    price: 79.99,
    category: 'Electronics',
    inStock: true,
    quantity: 15,
    rating: 4.5,
    imageUrl: 'assets/headphones.jpg',
    description: 'Premium wireless headphones with noise cancellation'
  },
  // Add at least 8 more products across different categories
  // Categories: Electronics, Clothing, Books, Home & Garden (you may use an AI to
  // generate this)
];
```

#### 3.2 Filter Functionality

- ☐ Dropdown to filter by category (All, Electronics, Clothing, Books, Home & Garden)
- ☐ Toggle to show only in-stock products
- ☐ Search input to filter by product name (real-time filtering)
- ☐ Display count of filtered products

#### 3.3 Sort Functionality

- ☐ Sort by: Price (Low to High), Price (High to Low), Name (A-Z), Rating
- ☐ Use ngModel for two-way binding with sort selection

#### 3.4 Display Logic

- ☐ Use \*ngFor to display product cards

- ☐ Use `*ngIf` to show "No products found" message when filter returns empty
- ☐ Show loading state initially (simulate with `setTimeout`)

## Styling Requirements

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### CSS Tasks:

- ☐ Create attractive product cards with hover effects
- ☐ Use CSS Grid or Flexbox for product layout
- ☐ Implement responsive design (mobile-friendly)
- ☐ Add transitions and animations for smooth interactions
- ☐ Use CSS classes for stock status (in-stock, low-stock, out-of-stock)
- ☐ Add visual feedback for button clicks and hovers